## Lecture 17

Let's consider an example problem

$$H(s) = \frac{2s+1}{s^2+2s+5}$$

Find the impulse response of this transfer function.

Let's modify the transfer function so that we can apply the standard Laplace transform.

$$H(s) = \frac{2s+1}{s^2+2s+5}$$

$$= \frac{2s+1}{s^2+2s+1+4}$$

$$= \frac{2s+1}{(s+1)^2 + 2^2}$$

We notice that, $\sigma = 1$, $\omega_d = 2$

so the two complex roots are

$$s = -\sigma \pm j\omega_d$$

$$s_1 = -1 + j2$$

$$s_2 = -1 - j2$$

now we modify the numerator

$$H(s) = \frac{2s+2-1}{(s+1)^2 + 2^2}$$

$$= 2 \cdot \frac{s+1}{(s+1)^2 + 2^2} - \frac{1}{(s+1)^2 + 2^2}$$

$$= 2 \cdot \frac{s+1}{(s+1)^2+2^2} - \frac{1}{2} \cdot \frac{2}{(s+1)^2+2^2}$$

This form is a standard form. We can apply the inverse Laplace transform to find the impulse response.

By looking at the relations from the previous lecture notes,

$$h(t) = 2e^{-t} \cdot \cos 2t - \frac{1}{2} e^{-t} \sin 2t$$

we found the impulse response of the transfer functions which has complex roots.